CS304P - Quiz No.04

Date: 08 January 2026

Q.1 Everything in LISP is considered as a _____.

A. Class
B. Function
C. Symbol
D. Object

Q.2 While eliminating repeated elements of list in PROLOG, each element should be replaced with:

A. Next element
B. Blank space
C. Single copy of the same element
D. Previous element

Q.3 How many times the value of a PROLOG variable can be changed after first assignment?

A. Many times
B. Once
C. Zero (Can not be changed)
D. Twice

Q.4 Which of the following are the types of constants in PROLOG?

A. float and numbers
B. short and char
C. atoms and numbers
D. single and decimal

Q.5 All the variables in PROLOG clauses/predicates are:

A. Strongly typed
B. Local
C. Global
D. Generic

Q.6 In a PROLOG program, the names of all relationships and objects must begin with:

A. Dollar sign ($)
B. Lowercase letter
C. Uppercase letter
D. Underscore

Q.7 Which of the following statement in LISP is used to list all properties of symbol 'k'?

A. >(get all 'k)

B. >(get 'k)

C. >(obtain all 'k)

D. >(symbol-plist 'k)

Q.8 The correct way to write Goal for following statement in PROLOG is _____. "Which medicine relieves in diarrhea?"

A. diarrhea (relieves, M).

B. relieves (M, diarrhea).

C. M (diarrhea, relieves).

D. M (relieves, diarrhea).

Q.9 Basic elements of a program in PROLOG are:

A. Entities and objects

B. Rules and queries

C. Facts and figures

D. Predicates and rules

Q.10 Following statement in PROLOG is considered as _____. "unsuitable_for(Person, Drug) :- aggravates(Drug, Condition), suffers_from(Person, Condition)."

A. Rule

B. Goal

C. Variable

D. Fact

Q.11 Which of the following function is used to add an element to the end of a vector?

A. back()

B. end()

C. insert()

D. push_back()

Q.12 How can you define a class template in C++?

A. template<class T> MyClass { ... };

B. template<type T> class MyClass { ... };

C. template<typename T> class MyClass { ... };

D. template<name T> class MyClass { ... };

Q.13 A class with pure virtual function is called _____ class.

A. normal

B. concrete

C. abstract

D. solid

Q.14 In case of public inheritance between Base class A and Derived class B, which of the folllowing data members of Base class will be hidden in Derived class?

A. protected

B. public

C. both public and private

D. private

Q.15 Which of the following is the correct syntax for creating a template class X friend of a template class Y?

A. friend X;

B. friend X<T>;

C. X<T>;

D. X<T> friend;

Q.16 There are _____ types of inheritance in C++.

A. 1

B. 4

C. 3

D. 2

Q.17 Which of the following is the correct syntax to declare a vector of type int?

A. sampleVector vector<int> ;

B. int <vector> sampleVector;

C. vector<int> sampleVector;

D. vector {int} sampleVector;

Q.18 Which of the following is not the correct type of inheritance in C++?

A. protected

B. secret

C. private

D. public

Q.19 If a function template has parameters `T` and `U`, calling `func(1, 2.5)` results in:

A. T=int, U=double

B. Both deduced as int

C. Compilation error

D. Both deduced as double

Q.20 What is the purpose of a generic algorithm in C++?

A. To make algorithms independent of data structures

B. To create run-time polymorphism

C. To avoid the use of pointers

D. To handle integer data only

Q.21 Virtual functions can be used in _____ ways.

A. 2

B. 8

C. 0

D. 6

Q.22 What is the purpose of explicit specialization in templates?

A. To improve runtime performance

B. To simplify the template syntax

C. To handle specific data types that the template cannot handle successfully

D. To avoid the use of templates

Q.23 If Vector<char* > is instantiated, which version will the compiler choose?

A. Partial specialization

B. General template

C. Complete specialization

D. Function overloading

Q.24 In a function template template<typename T>, what does T represent?

A. Type parameter

B. Function return value

C. Non-type parameter

D. Built-in type

Q.25 Which one of the following is an example of complete specialization?

A. template <class T> class Vector<T *> { }

B. template <typename T> class Vector<T*> { }

C. template<> class Vector <char*> { }

D. template <typename T> class Vector { }

Q.26 Which keyword is used to define a template in C++?

A. function

B. class

C. typename

D. template

Q.27 class A { public: int Test(int x) {return x*x;} int Test(int x,int y) {return x*y;} }; class B : public A { public: int Test(int x) { return x*x*x;} }; In the above class, the method Test is_____.

A. Overridden

B. None of the given options

C. Both

D. Overloaded

Q.28 Which of the following instantiates a template with explicit type?

A. typename max(a,b)

B. template max(a,b)

C. max<T>(a, b)

D. max(a, b)

Q.29 Function templates are preferred over overloading when

A. They have different numbers of arguments

B. Functions differ only by parameter types

C. They return different types

D. They have different names

Q.30 Which of the following is the correct syntax for creating a function Test(T) friend of a template class?

A. void friend Test(T);

B. void Test(T);

C. void Test(T) friend;

D. friend void Test(T);
